Noun-Noun and Adjective-Noun Combinations in Computer Science Journal Writings

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Introduction

Computer Science is a highly technical and informative field that contains an abundant number of pre-modified nouns, which help provide readers with a clearer understanding of the topics to be discussed. Two common types of pre-modified nouns found in Computer Science journals are noun-noun combinations and adjective-noun combinations and are used to show complexity and advancement to the reader (Wang, Ge, & Ren, 2022). Several abstracts from the *Science Publications - Journal of Computer Science* will be studied and analyzed, focusing on the number of noun-noun and adjective-noun combinations presented only in disciplinary terminology. In general, technical based writing needs to be incredibly descriptive, and having pre-modified nouns allows for grammar that is more descriptive to enhance the reader's understanding (Spangler, 1956). This analysis will demonstrate how pre-modified nouns contribute to the clarity, precision, and depth of technical writing in the field of Computer Science by underscoring their key role in communicating complex ideas and advancing disciplinary understanding.

Typically, in these two types of combinations, the second (or only) noun is considered the head noun and indicates how it differs from other categories (Raffray, Pickering, & Branigan, 2007). A noun-noun combination involves two or more nouns functioning together as a single unit to convey specific information about the head noun. In contrast, an adjective-noun combination features an adjective placed before a noun to describe or refine its meaning. This grammatical structure helps convey meaning or adds situational context to the noun, enhancing its interpretation (Murphy, 1990). Draskovic, Pustejovsky, and Schreuder (2013) examine the distinction between what this paper will refer to as discipline-specific terminology versus non-discipline-specific terminology. In this research, there will be a specific distinction between the two forms of terminology and which will be quantified.

Method

The research process involves analyzing fifteen abstracts from journal articles published in a computer science focused journal. The selected source is *Science Publications, Journal of Computer Science*, from which the abstracts will be examined. This journal requires abstracts to be under three hundred words and to clearly communicate the focus of the article. All selected entries were published within the last four years, relative to 2025, were peer-reviewed and approved by the journal, and demonstrated a strong emphasis on topics such as algorithms, machine learning, or computer science development. Computer Science is an incredibly broad field, so to ensure clarity and focus, the discipline has been divided into more specific subfields. Titles or abstracts must include at least one of the following keywords: Database, Artificial Intelligence (AI), Algorithms, Encryption, Mining Techniques, Parsing Tools, Machine Learning (e.g., Classification Mode, Attention Mode), Phishing, or Cryptography.

Combinations found in the writings were divided into six categories: Discipline Specific Noun-Noun Combinations, Non-Discipline Specific Noun-Noun Combinations, Discipline Specific Adjective-Noun Combinations, Non-Discipline Specific Adjective-Noun Combinations, Discipline Specific Combination of Both, and Non-Discipline Specific Combination of Both. Each phrasal combination that is placed in the discipline specific category must be meaningful and contextually relevant to computer science, ensuring that it enhances the understanding of the subject matter for readers within the field. Once the analysis is complete, the frequency of noun-noun and adjective-noun combinations will be counted and compared to assess the differences.

When examining word combinations in technical and academic writing, two common structures emerge: noun-noun and adjective-noun combinations. A noun-noun combination involves two or more nouns working together as a single unit to convey specific information about the head noun. For example, in the discipline-specific term *Database Industry*, *Industry* is the head noun, while *Database* provides additional detail—both are nouns. Similarly, in *Computer Science*, *Science* is the head noun modified by the noun *Computer*. Even outside of discipline-specific contexts, noun-noun pairings like *Analysis Process* follow the same pattern, with *Process* as the head noun. In contrast, an adjective-noun combination places an adjective before a noun to refine or describe its meaning. For instance, in *Representative Algorithms*, *Algorithms* is the head noun, and *Representative* provides descriptive context. Other examples include *Labeled Datasets* (discipline-specific) and *Guided Solution* (non-discipline-specific). More complex combinations can involve additional elements. A term like *Software Diagnostic System* is a noun-noun-noun combination, while *Efficient Data Encryption* is an adjective-noun-noun combination, where *Efficient* modifies the entire noun phrase *Data Encryption*. Interestingly, *Efficient Data Encryption* appears in both adjective-noun and noun-noun classifications, highlighting the layered structure often found in technical language. In cases where a phrase can reasonably fall into both the noun noun and adjective noun categories, such as *Efficient Data Encryption*, the phrase will be classified under a separate category titled Discipline or Non Discipline Specific Combination of Both. This ensures clarity and avoids redundancy in categorization.

Analyses

These journals were selected on the basis of relevance to the target academic discipline and their status as both seminal and current research. The data set comprises 3,104 words of text, from the abstracts of the fifteen articles. This information was analyzed for two types of lexical structures: noun-noun combinations and adjective-noun combinations. Analysis was manual, with the researcher's knowledgeable understanding of discipline-specific and non-discipline-specific lexicon informing identification and categorization of each instance.

One of the first significant findings from the data is the proportion of discipline-specific to non-discipline-specific words. Of all the noun-noun combinations that were identified in the abstracts, 60.47% were discipline-specific, and 39.54% were non-discipline-specific. This suggests that noun-noun combinations are more likely to convey technical or specialized concepts that are closely associated with the discourse community of the discipline. In comparison, adjective-noun combinations not only appeared more often but also switched in the instances shown in discipline versus non-discipline specific. That is, 80.30% of the adjective-noun combinations were non-discipline-specific and a mere 19.70% were discipline-specific. This may indicate that adjective-noun combinations are more linguistically functional and general, bridging technical terminology and more general academic terminology.

In regards to the complex combinations, they are the least represented instances in the abstracts, but Discipline Specific Combinations of Both show prevalence over the Non-Discipline Specific Combinations of both by 8.75% of the 423 total instances. Of all the combinations, the Combinations of Both only make up 12.53% of all of the phrases present.

Table 1

Total Combination Count among Categories

Total Combination Count Total Combinations: 423 Instances					
Total Noun-Noun Combinations	**Discipline Specific Noun-Noun Combinations**	**Non-Discipline Specific Noun-Noun Combinations**	**Total Adjective-Noun Combinations**	**Discipline Specific Adjective-Noun Combinations**	**Non-Discipline Specific Adjective-Noun Combinations**
172 Instances	104 Instances	68 Instances	198 Instances	39 Instances	159 Instances
Discipline Specific Combination of Both			**Non-Discipline Specific Combination of Both**		
37 Instances			16 Instances		

Although noun–noun phrases are typically more prevalent in discipline-specific writing, Abstract Six presents a notable exception: it contains only two noun–noun combinations compared to eight adjective–noun combinations. This deviation is particularly noteworthy because, while uncommon, such variations do occur and highlight the stylistic flexibility that can exist within academic writing. Recognizing these outliers is important, as they can offer insight into the author's rhetorical choices and the emphasis placed on descriptive precision. Refer to Table 2 to show the representation of a full breakdown of all of the combinations in each abstract. Refer to Table 3 to see a broad overview of the analysis.

Table 2

Abstract Specific Combination Analysis

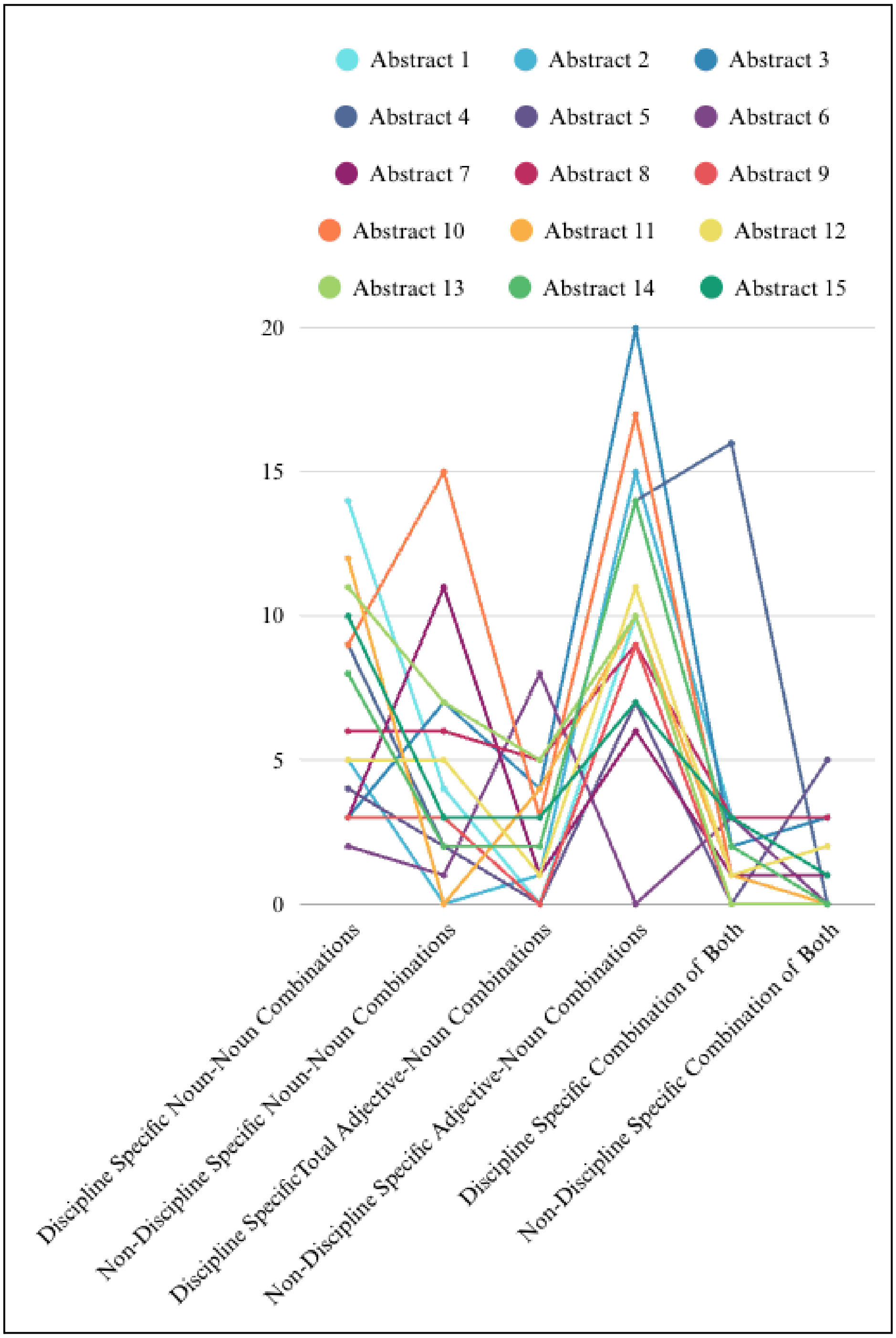
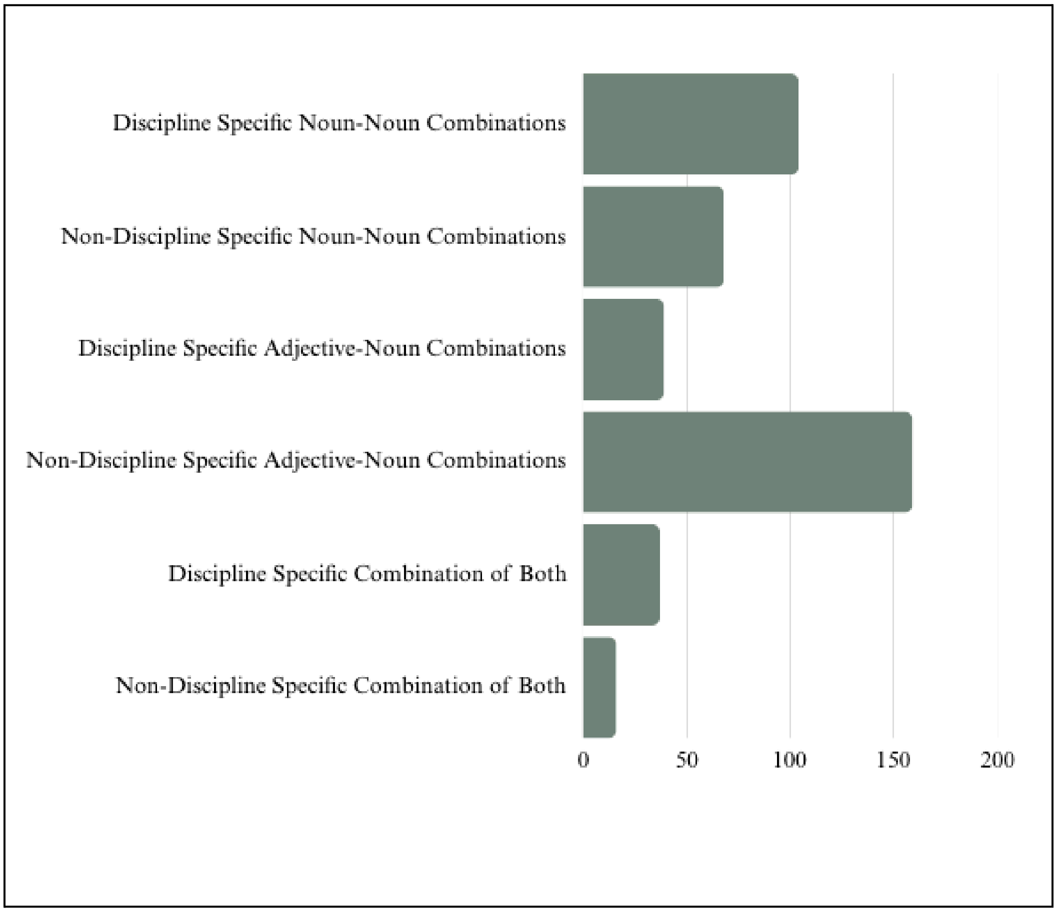


Table 3

Broad Overview of Combinations Analysis



These results suggest an incipient pattern of lexical use in academic writing in the discipline. Noun-noun combinations appear to have a more precise, field-specific communicative function, while adjective-noun combinations allow for greater descriptive latitude. Identification of the pattern and application of these lexical pairs can inform subsequent knowledge of the conventions of academic writing and assist in guiding future work in genre-sensitive language instruction or corpus-based inquiry. Further research on a larger data set will be required to confirm whether these trends are accurate across a broader spectrum of texts.

Discussion

In total, 423 phrasal combinations were identified and classified into six categories: discipline-specific and non-discipline-specific noun–noun combinations, discipline-specific and non-discipline-specific adjective–noun combinations, and combinations with elements of both. Of the 172 noun–noun combinations, 60.47% were discipline-specific, justifying their utility in conveying technical concepts such as Database Industry and Computer Science. Such combinations often occur as effective representations of technical information that instantaneously express themselves within the discourse community.

On the other hand, adjective–noun pairs were more frequent overall (198), but were predominantly non-discipline-specific (80.30%). This division shows that while these pairs are not as field-specific in their vocabulary, they play a vital role in fulfilling the descriptive and explanatory role of abstract writing. Phrases like Guided Solution and Representative Algorithms reveal this trend, giving a more flexible linguistic expression better adapted to highlighting features, functions, or characteristics.

The study also addressed compound combinations, phrases that might reasonably fall under both categories since they have a stratified structure (e.g., Efficient Data Encryption). Even though these combinations made up just 12.53% of the total, the discipline-specific variants were 8.75% more than their non-discipline-specific versions, further supporting the observation that technical language has a preference for compound precision.

Exceptions like Abstract Six, in which adjective noun combinations occur exponentially more than noun noun combinations, demonstrate the heterogeneity present in academic language. Such deviations demonstrate that author intention, stylistic preference, or a study's specific focus can govern lexical choice, undermining assumptions of homogeneity in technical writing.

Overall, these results indicate an emerging trend in computer science academic language use. Noun noun compounds fulfill the disciplinary specificity and technical precision role, while adjective noun combinations are capable of yielding descriptive flexibility and broad academic use. The use of both forms in collocation can offer a touchstone for writers wishing to write suitably in the field. Further, the simple separation of discipline specific and non discipline specific pairings provides an applied model of genre sensitive pedagogy and corpus driven analysis. In order to validate these findings, additional research would require the expansion of the dataset to examine if these patterns are consistent across a broader body of technical writing.

Conclusion

This study compared noun-noun and adjective-noun combinations within Journal of Computer Science abstracts to examine the manner in which lexical structures function in disciplinary writing. The findings show a highly skewed preference toward discipline-specific terminology in noun-noun combinations, 91.67% of which are defined as such. This indicates that these structures must be used in order to report technical specificity and convey specialist thinking within the field. On the other hand, combinations of adjective-nouns were nearly evenly distributed between discipline usage (53.23%) and non-discipline usage (46.77%), reflecting that they serve a more generic descriptive purpose in academic texts.

The contrast between these two structures suggests the mechanism by which academic writers negotiate specificity against greater access. Noun-noun compounds contribute to the formal, discipline-specific nature of computer science prose, while adjective-noun compounds offer flexibility for contextual or explanatory deployment. Knowledge of the role of these structures can be used to support the teaching of academic writing, particularly in genre-sensitive language teaching and corpus-driven instruction.

Although these preliminary findings tell us something about the language inclinations of three computer science abstracts, the sample is small and restricted in its scope. More research with a representative and larger sample would inform us as to whether patterns are persistent within subfields and publication type. This would inform us further regarding the nature of language used in technical writing and influence instructional and communication best practice within the discipline.

Reflection

I have a BA in Computer Science with a minor in Mathematics. I am also currently earning my master's at ODU in English, with a focus in technical writing. Career-wise, I currently work in the STEM field as a technical writer and a C#/Unity programmer. I have spent the last seven years of my life working in STEM, and I thought this would be a great topic to delve into. I honestly really enjoyed this research, and as far as my technical writing background goes, it was extremely valuable. In this class and with this paper, grammar tools and structures are obviously everywhere, but this is the first time I truly paid them any attention. This is so important in technical writing, given the heavy communication we have with engineers and the challenge of conveying their speech in ways that users can understand. I found this especially prevalent in these extremely technical abstracts. I enjoyed spending time with this subject and learning more about a side of the field I haven't explored much before.

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Draskovic, I., Pustejovsky, J., & Schreuder, R. (2013). Adjective-noun combinations and the generative lexicon. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society* (pp. 181–202). https://doi.org/10.1007/978-94-007-5189-7

Murphy, G. L. (1990). Noun phrase interpretation and conceptual combination. *Journal of Memory and Language, 29*(3), 259–288. https://doi.org/10.1016/0749-596X(90)90001-G

Wang, Y., Ge, T., & Ren, Z. (2022). Noun phrasal complexity in computer science conference abstracts: A corpus-based study. *International Journal of Computer-Assisted Language Learning and Teaching, 12*, 1–17. https://doi.org/10.4018/IJCALLT.311096

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Spangler, E. R. (1956). Modern grammar and its application to technical writing. *Journal of Chemical Education, 33*(2), 61. https://doi.org/10.1021/ed033p61

Modern Grammar and its Application to Technical Writing, written by E. R. Spangler discusses the typical grammatical application in technical writing. The nature of writing in technical fields requires incredibly descriptive language, and this journal discusses those regulations, and common mishaps writers run into. This work offers valuable insight into how technical communication is enhanced by grammatical accuracy, and therefore it is particularly relevant to studies of linguistic practice in scientific and academic writing.

Raffray, C. N., Pickering, M. J., & Branigan, H. P. (2007). Priming the interpretation of noun–noun combinations. *Journal of Memory and Language, 57*(3), 380–395. https://doi.org/10.1016/j.jml.2007.06.009

Priming the Interpretation of Noun-Noun Combinations, written by Claudine N. Raffray, Martin J. Pickering, and Holly P. Branigan, gives an overview of how readers interpret noun-noun combinations, specifically focusing on the effects of lexical repetition and semantic relation. This discussion advances the researchers' understanding of how noun-noun combinations are perceived and analyzed.

Draskovic, I., Pustejovsky, J., & Schreuder, R. (2013). Adjective-noun combinations and the generative lexicon. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society* (pp. 181–202). https://doi.org/10.1007/978-94-007-5189-7

Adjective-Noun Combinations and the Generative Lexicon, written by Irena Drasković, James Pustejovsky, and Rob Schreuder, delves into adjective-noun combinations and their computational complexity and conceptual compatibility. This analysis is crucial in understanding the significance of adjective-noun combinations, discipline-specific complexity often present in technology-based academic journals and technical writings across various subfields.

Murphy, G. L. (1990). Noun phrase interpretation and conceptual combination. *Journal of Memory and Language, 29*(3), 259–288. https://doi.org/10.1016/0749-596X(90)90001-G

Noun Phrase Interpretation and Conceptual Combination, written by Gregory L. Murphy, discusses how both noun-noun combinations and adjective-noun combinations are perceived in writing and whether readers find them comprehensible. In addition, the article explores which of the two combinations is more easily understood by readers and how each contributes to grasping the overall context of the message being conveyed.

Wang, Y., Ge, T., & Ren, Z. (2022). Noun phrasal complexity in computer science conference abstracts: A corpus-based study. *International Journal of Computer-Assisted Language Learning and Teaching, 12*, 1–17. https://doi.org/10.4018/IJCALLT.311096

Noun Phrasal Complexity in Computer Science Conference Abstracts: A Corpus-Based Study, written by Yu Wang, Tianshuang Ge, and Zhilei Ren, analyzes the presence of noun phrases in computer science abstracts. The study examines the quantification of noun phrases and identifies which pre-modifiers of nouns are typically preferred by the authors of the academic journals.

Journals Investigated

1. Alraddadi, A. S. (2024). Reconstruction investigation model for database management systems. *Journal of Computer Science, 20*(1), 33–43. https://doi.org/10.3844/jcssp.2024.33.43

2. Alshammari, B. M. (2024). COVID-19 in the era of artificial intelligence: Existing technologies and a strategic model for mitigating future pandemics. *Journal of Computer Science, 20*(5), 465–486. https://doi.org/10.3844/jcssp.2024.465.486

3. Aslanov, T. G., Ivanov, N. Z., Anudinov, S. S., & Karlova, T. V. (2025). Algorithmization of electrostatic model for solving the direct problem of electrography. *Journal of Computer Science, 21*(4), 869–882. https://doi.org/10.3844/jcssp.2025.869.882

4. Awasthi, S., & Kohli, N. (2025). Efficient data encryption for securing HDFS using DQN-enhanced deep reinforcement learning techniques. *Journal of Computer Science, 21*(4), 741–760. https://doi.org/10.3844/jcssp.2025.741.760

5. Geasela, Y. M., Bernanda, D. Y., Andry, J. F., Jusuf, C. K., Winata, S., Lydia, & Everlin, S. (2024). Analysis of student mental health dataset using mining techniques. *Journal of Computer Science, 20*(1), 121–128. https://doi.org/10.3844/jcssp.2024.121.128

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elliptic curve cryptography. *Journal of Computer Science, 18*(5), 322–338. https://doi.org/10.3844/jcssp.2022.322.338

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9. Gupta, S., Adhikari, U., Varshney, S., & Choudhury, T. (2025). Computational analysis of genre effects on movie ratings using MLP algorithms. *Journal of Computer Science, 21*(4), 905–917. https://doi.org/10.3844/jcssp.2025.905.917

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Appendix

Abstract Analysis

Note: Bolded text is Noun-Phrase Combinations and underlined text is Adjective-Noun Combinations. Red highlighted content is non-discipline specific and Green highlighted content is discipline specific.

1. Alraddadi, A. S. (2024). Reconstruction investigation model for database management systems. *Journal of Computer Science, 20*(1), 33–43. https://doi.org/10.3844/jcssp.2024.33.43

There have been <u>increased levels</u> of cybercrime in the **database industry**, which has hurt the confidentiality, integrity, and availability of these systems. Most organizations apply several **security layers** to detect and prevent **database crimes**. For this reason, **Database Forensics** (DBF) plays a very <u>important role</u> in capturing and discovering who the criminal is, when the crime was committed, and which part of the database the crime occurred. Several <u>forensic models</u> have been proposed for the **DBF field**, which can be used to identify, collect, preserve, examine, analyze, and document **database crimes**. However, most of these models focused on <u>specific **database systems**</u> due to the variety of the **database infrastructure** and the <u>multidimensional nature</u> of the **database systems**. The most <u>important part</u> of the **DBF field** is the **analysis process** used to investigate the <u>captured data</u> and discover the attack. Thus, this study proposes an **Integrated Reconstruction Investigation Model** (IRIM) for **database forensics** using a **metamodeling method**. It consists of two <u>main processes</u>: The <u>examining process</u> and the <u>discovering and reporting process</u>. A <u>real scenario</u> has been used to validate the effectiveness of the **proposed model**. According to the results, the **proposed model** could detect

database cybercrimes and allow **domain forensic practitioners** to capture and analyze **database crimes** efficiently.

Word Count: 208

Total Combinations: 29

Total Noun-Noun Combination: 18

Noun-Noun Combination (Discipline Specific): 14

Noun-Noun Combination (Non-Discipline Specific): 4

Total Adjective-Noun Combination: 10

Adjective-Noun Combination (Discipline Specific): 0

Adjective-Noun Combination (Non-Discipline Specific): 10

Combination of Both: 1 (DS) 0 (NDS)

2. Alshammari, B. M. (2024). COVID-19 in the era of artificial intelligence: Existing technologies and a strategic model for mitigating future pandemics. *Journal of Computer Science, 20*(5), 465–486. https://doi.org/10.3844/jcssp.2024.465.486

Pandemics have existed since the existence of life and will continue as life continues. Throughout many of the previous pandemics, what played a major role in decreasing their severity is how we mitigated and controlled them. The main reason for this is the time it takes for treatments and vaccinations to be developed, which usually takes a long time. Therefore, the techniques used to control a pandemic rapidly change over the course of the pandemic until a cure or vaccine comes to light. At present, emerging technologies such as **Artificial Intelligence** (AI), the Internet of Thing**s** (IoT), **fifth generation networks** and big data can have a significant impact on controlling upcoming pandemics including. This study provides a comprehensive survey of current technologies that use **AI** and big **data analytics** to take part in the fight against the current pandemic (COVID-19), including their objectives, strengths, weaknesses and challenges. This study also studies existing **telemedicine technologies** and **contact tracing tools** used in various countries, which governments have adapted to fight against the current **COVID-19 pandemic**. This study concludes by suggesting a novel strategic model for controlling and mitigating pandemic crises (e.g., COVID-19). This model represents a guided solution for identifying pandemics and for controlling them using advanced **digital solutions** from the early stages. More precisely, it can assist in identifying which **AI tools** can be developed in order to predict pandemics before their outbreaks.

Word Count: 233

Total Combinations: 25

Total Noun-Noun Combination: 5

Noun-Noun Combination (Discipline Specific): 5

Noun-Noun Combination (Non-Discipline Specific): 0

Total Adjective-Noun Combination: 16

Adjective-Noun Combination (Discipline Specific): 1

Adjective-Noun Combination (Non-Discipline Specific): 15

Combination of Both: 3 (DS) 1(NDS)

3. Aslanov, T. G., Ivanov, N. Z., Anudinov, S. S., & Karlova, T. V. (2025). Algorithmization of electrostatic model for solving the direct problem of electrography. *Journal of Computer Science, 21*(4), 869–882. https://doi.org/10.3844/jcssp.2025.869.882

An analysis of the creation and implementation of algorithms to control cardiological processes occurring in the **cardiovascular system** using **software and hardware diagnostic systems** is a required activity. This study considers ways to improve the methods of medica**l data processing** using **electronic diagnostic systems**, relying on existing methods of mathematical simulation of **human body states**. This study describes the development of a method that solves the direct problem of electrical cardiography by using complete **electrostatic simulations**. For this purpose, the selection of key characteristics of simulations of anatomical features and physiology of an organism is used by considering the bioelectrical data obtained experimentally. This study describes the basic relations to characterize the **electric field** and considers the electrophysical properties of biological tissues. The simulation considered the specifics of the functioning of the proposed method, which imitates the activity of the **heart muscle**. A novel algorithm is proposed that allows for the detection of myocardial pathological changes that arise during **diagnostic procedures**, utilizing the reconstructed trajectory of single **charge movement**. An in-depth study of the electric fields of the controlled object using **multipole dispersions** was conducted, and the impact of the level of **multipole dispersions** on the reliability and error of the **simulation results** was analyzed. The proposed simulation does improve the reliability of the parameters obtained during diagnosis using electronic **cardiac systems**. The implementation of **calculation methods** for electrophysiological signals based on this model has the potential to facilitate diagnostic procedures even in the presence of undetected pathologies, offering a valuable complement to traditional **diagnostic techniques**.

Word Count: 256

Total Combinations: 39

Total Noun-Noun Combination: 10

Noun-Noun Combination (Discipline Specific): 3

Noun-Noun Combination (Non-Discipline Specific): 7

Total Adjective-Noun Combination: 24

Adjective-Noun Combination (Discipline Specific): 4

Adjective-Noun Combination (Non-Discipline Specific): 20

Combination of Both: 2 (DS) 3 (NDS)

4. Awasthi, S., & Kohli, N. (2025). Efficient data encryption for securing HDFS using DQN-enhanced deep reinforcement learning techniques. Journal of Computer Science, 21(4), 741–760. https://doi.org/10.3844/jcssp.2025.741.760

In the era of big data, ensuring the security of large-scale **distributed storage systems** like the **Hadoop Distributed File System** (HDFS) is critical. Traditional **encryption methods** often struggle to balance robust security with **system performance**, leading to vulnerabilities and inefficiencies. This study presents the design and implementation of an efficient **data encryption algorithm** for securing HDFS using **Deep Q-Network** (DQN) enhanced **Deep Reinforcement Learning** (DRL) techniques. The proposed model dynamically optimizes **encryption parameters** by leveraging the adaptive capabilities of DQN, ensuring robust security while maintaining high **system performance** and scalability. Our approach addresses key limitations of existing **encryption methods** by integrating DQN with **deep reinforcement learning** to create a dynamic **encryption framework** that adjusts in real time based on **data access patterns** and **threat levels**. The results demonstrate significant improvements in security and efficiency compared to conventional **encryption techniques**. Specifically, the DQN-enhanced **DRL**

algorithm consistently outperformed baseline methods regarding **encryption strength**, computational efficiency, latency, **resource utilization**, adaptability, and **energy consumption**. The contributions of this research include the development of a novel DQN-enhanced **encryption algorithm** tailored for HDFS, the creation of an adaptive **encryption framework** that leverages real-time **data dynamics**, and a thorough evaluation demonstrating the practical benefits of the proposed solution. This study paves the way for future research in intelligent **encryption systems**, offering a robust and efficient approach to securing large-scale **distributed storage environments**. Our findings underscore the potential of integrating advanced **machine learning techniques** into **encryption processes** to enhance security and performance, addressing the complex challenges modern **data storage systems** pose.

Word Count:253

Total Combinations: 43

Total Noun-Noun Combination: 11

Noun-Noun Combination (Discipline Specific): 9

Noun-Noun Combination (Non-Discipline Specific): 2

Total Adjective-Noun Combination: 16

Adjective-Noun Combination (Discipline Specific): 2

Adjective-Noun Combination (Non-Discipline Specific): 14

Combination of Both: 16 (DS) 0 (NDS)

5. Geasela, Y. M., Bernanda, D. Y., Andry, J. F., Jusuf, C. K., Winata, S., Lydia, & Everlin, S. (2024). Analysis of student mental health dataset using mining techniques. Journal of Computer Science, 20(1), 121–128. https://doi.org/10.3844/jcssp.2024.121.128

This study utilizes a **decision tree model** in RapidMiner to analyze a dataset from Kaggle, comprising 200 **student records**. Among these, 70 students reported mental **health issues**, while 130 did not. Strikingly, a significant majority of 58 out of the 70 students with mental **health concerns** do not seek assistance from professionals. This study underscores the pressing issue of underutilization of mental **health services** among students and offers practical solutions, such as enhancing awareness and education, improving access to mental **health services,** providing **peer support**, and addressing underlying issues. The **research design** includes **data collection methods** that maintained ethical standards and the **decision tree model'**s application for analysis. This study's contribution lies in its identification of the prevalence of students with mental **health issues** who do not seek help and the proposed solutions to address this critical issue.

Word Count: 138

Total Combinations: 18

Total Noun-Noun Combination: 6

Noun-Noun Combination (Discipline Specific): 4

Noun-Noun Combination (Non-Discipline Specific): 2

Total Adjective-Noun Combination: 7

Adjective-Noun Combination (Discipline Specific): 0

Adjective-Noun Combination (Non-Discipline Specific): 7

Combination of Both: 0 (DS) 5 (NDS)

6. Ganda, Y. M., Naroua, H., & Idi, B. M. (2025). A generic syntax analyzer for the computerization of African languages. Journal of Computer Science, 21(4), 810–816. https://doi.org/10.3844/jcssp.2025.810.816

This article enters within the framework of a contribution to the computerization of African Languages. It presents a generic **parsing tool** designed for West African languages despite their individual specificities. The genericity of the tool resides in the fact that it is adaptable to any West African language while using a minimum of resources. It only needs an **etiquette dictionary** and context-free grammar. From these two resources, the tool is designed to generate an **LR (1) parser**. The choice of this technique is based on its supremacy over other techniques despite its disadvantage of generating an exponentially large **parsing table**. However, to reduce the size of the automaton while maintaining the power of the technique, the system brings together dictionary entries depending on their type, gender, and number in the form of terminals. The results obtained show that it is possible to generate a relatively small and efficient **analysis table** instead of a large one. An application of the proposed tool to the **Hausa nominal phrases** has succeeded in reducing the number of terminals from more than 10,000 to only 88. Instead of having an automaton with thousands of states, we end up with an automaton with only 198 states.

Word Count: 201

Total Combinations: 14

Total Noun-Noun Combination: 3

Noun-Noun Combination (Discipline Specific): 2

Noun-Noun Combination (Non-Discipline Specific): 1

Total Adjective-Noun Combination: 8

Adjective-Noun Combination (Discipline Specific): 8

Adjective-Noun Combination (Non-Discipline Specific): 0

Combination of Both: 3 (DS) 0 (NDS)

7. Gupta, N., Gupta, B., Passi, K., & Jain, C. K. (2022). Applications of artificial intelligence based technologies in weed and pest detection. Journal of Computer Science, 18(6), 520–529. https://doi.org/10.3844/jcssp.2022.520.529

Unprecedented **population growth** and **climate change** has burdened **food security** and scarcity worldwide, where the **agriculture sector** can significantly contribute to accomplishing the demands and contribute to the **economic growth** of a country. **Artificial Intelligence** (AI) has revolutionized the agricultural domain. **Pest and weed detection** is significant to yielding good quality crops. The AI-based **tools and technologies** such as drones and robots bring advancement in **crop production** by performing the early detection of weeds and pests. The tools utilize **image processing** and **machine learning algorithms** to capture, analyze and detect the presence of weeds and pests in plants. The **research work** carried out provides a comprehensive survey for the application of **artificial intelligence** for both **weed and pest detection**. It presents widely used techniques, their **evaluation parameters**, and publicly available datasets which provide the current status of work for the researchers working in the domain of **weed and pest detection**.

Word Count: 150

Total Combination: 23

Total Noun-Noun Combination: 14

Noun-Noun Combination (Discipline Specific): 3

Noun-Noun Combination (Non-Discipline Specific): 11

Total Adjective-Noun Combination: 7

Adjective-Noun Combination (Discipline Specific): 1

Adjective-Noun Combination (Non-Discipline Specific): 6

Combination of Both: 1 (DS) 1 (NDS)

8. Gupta, S., Adhikari, U., Varshney, S., & Choudhury, T. (2025). Computational analysis of genre effects on movie ratings using MLP algorithms. Journal of Computer Science, 21(4), 905–917. https://doi.org/10.3844/jcssp.2025.905.917

Predictive analytics are what the **entertainment industry** depends on so much in prognosticating **movie ratings**, which is why they inform filmmakers, distributors and **stream platforms** strategic moves. Traditional **prediction models** most times only succeed in missing out on the intricate dynamics that are genre-specific to **movie ratings** thereby leading to inaccuracies and suboptimal **decision making**. The paper proposes this research presents a comprehensive mechanism that combines extensive metadata with **Multi-Layer Perceptron (MLP) models** to increase the accuracy of predictions across multiple **cinematic genres**. Therefore, we had a goal of establishing fine patterns due to **MLP regressors** based on specific genres as well as addressing limitations linked to traditional approaches. To conduct this study; we used **principal component analysis** and **one-hot encoding** for 950 films followed by genre-specific modeling alongside statistical tests such as ANOVA, t-tests, **Gradient Boosting Classifiers** among others for **model validation**. They found that **adventure movies** were more predictable than other genres (MSE = 0.023) such as action (MSE = 2.816). It's clear then that accurate modelling requires an examination by gender and broad **data sources** integration. The research emphasizes the potential of improved **machine learning methods** to change predictive modeling in the area of art. Further work will seek to develop more accurate **feature selection**, deal with **data imbalance** and incorporate real-time **audience engagement** measures into the **optimization process** for better predictions that would help **film makers** make better strategic decisions.

Word Count: 234

Total Combination: 32

Total Noun-Noun Combination: 12

Noun-Noun Combination (Discipline Specific): 6

Noun-Noun Combination (Non-Discipline Specific): 6

Total Adjective-Noun Combination: 14

Adjective-Noun Combination (Discipline Specific): 5

Adjective-Noun Combination (Non-Discipline Specific): 9

Combination of Both: 3 (DS) 3 (NDS)

9. Lopez-Zevallos, L., Navarro-Prado, E., & Cabanillas-Carbonell, M. (2025). Virtual learning system with artificial intelligence for the academic management process. Journal of Computer Science, 21(4), 827–835. https://doi.org/10.3844/jcssp.2025.827.835

The impact of the **COVID-19 pandemic** on global education, with the massive closure of schools and universities. It highlights the challenge of adapting to **distance learning modalities** and the loss of **students' knowledge**. In this context, the use of **Artificial Intelligence** (AI) has become a promising alternative to improve distance education; it is presented as a solution to automate administrative tasks and personalize learning. The research work conducted focuses on how an **AI-based e-learning system** has improved **academic management** and learning at the institution. The study involved the participation of 20 experts, who evaluated the prototype in terms of efficiency, usability, technology, functionality, and design. As a result of the evaluation of the criteria, 93% in design and 92% in functionality were obtained. Likewise, 68.9% of experts determined that the system has a "Good" efficiency, with the highest percentage followed by 64.4% of "very good" efficiency. In conclusion, efficiency has a high degree of satisfaction due to being fast and efficient.

Word Count: 163

Total Combinations: 15

Total Noun-Noun Combination: 6

Noun-Noun Combination (Discipline Specific): 3

Noun-Noun Combination (Non-Discipline Specific): 3

Total Adjective-Noun Combination: 9

Adjective-Noun Combination (Discipline Specific): 0

Adjective-Noun Combination (Non-Discipline Specific): 9

Combination of Both: 0 (DS) 0 (NDS)

10. Nagendra, N., & Chandra, J. (2024). Sentence classification using attention model for e-commerce product review. Journal of Computer Science, 20(5), 535–547. https://doi.org/10.3844/jcssp.2024.535.547

The importance of **aspect extraction** in **text classification**, particularly in the **e-commerce sector**. **E-commerce platforms** generate vast amounts of **textual data**, such as comments, **product descriptions**, and **customer reviews**, which contain valuable information about various aspects of products or services. **Aspect extraction** involves identifying and classifying individual traits or aspects mentioned in **textual reviews** to understand **customer opinions**, improve products, and enhance the **customer experience**. The role of **product reviews** in e-commerce is discussed, emphasizing their value in aiding **customers' purchase decisions** and guiding businesses in **product stocking** and **marketing strategies**. Reviews are essential for boosting sales potential, maintaining a good reputation, and promoting **brand recognition**. Customers extensively research **product reviews** from different sources before purchasing, making them vital user-generated content for **e-commerce businesses**. The current work provided an efficient and novel **classification model** for **sentence classification** using the **ABNAM model**. The automated text **classification models** available cannot categorize the data into sixteen distinct classes. The technologies applied for the mentioned work contain TF-IDF, N-gram, CNN, linear SVM, **random forest**, **Naïve bays**, and ABNAM with significant results. The best-performing **ML method** for the successful classification of a given sentence into one of the sixteen categories is achieved with the proposed model named the **based Neural Attention Model** (ABNAM), which has the highest accuracy at 97%. The research acclaimed ABNAM as a novel classification model with the highest-class categorizations.

Word Count: 228

Total Combinations: 45

Total Noun-Noun Combination: 24

Noun-Noun Combination (Discipline Specific): 9

Noun-Noun Combination (Non-Discipline Specific): 15

Total Adjective-Noun Combination: 20

Adjective-Noun Combination (Discipline Specific): 3

Adjective-Noun Combination (Non-Discipline Specific): 17

Combination of Both: 1 (DS) 0 (NDS)

11. Nagunwa, T. (2024). Detection of phishing websites hosted in name server flux networks using machine learning. Journal of Computer Science, 20(1), 10–32. https://doi.org/10.3844/jcssp.2024.10.32

Attackers are increasingly using **Name Server IP Flux Networks** (NSIFNs) to run the **domain name services** of their **phishing websites** in order to extend the duration of their phishing operations. These networks host a **name server** that manages the **Domain Name System (DNS) records** of the websites on a network of compromised machines with frequently changing **IP addresses**. As a result, blacklisting the machines has less impact on stopping the services, lengthening their lifespan and that of the websites they support. High detection delays and the use of fewer, lesser varied **detection features** limit the proposed solutions for identifying the websites hosted in these networks, making them more susceptible to **detection evasions.** This study suggests a novel set of highly diverse features based on **DNS, network, and host behaviors** for fast and highly accurate detection of **phishing websites** hosted in NSIFNs using a

Machine Learning (ML) **approach**. Using a variety of traditional and deep learning **ML algorithms**, the prediction performance of our features was assessed in the context of binary and multi-class classification tasks. Our approach achieved optimal accuracy rates of 98.59% and 90.41% for the binary and multi-class classification tasks, respectively. Our approach is a crucial step toward monitoring **NSIFN components** to mitigate phishing attacks efficiently.

Word Count: 209

Total Combinations: 27

Total Noun-Noun Combination: 12

Noun-Noun Combination (Discipline Specific): 12

Noun-Noun Combination (Non-Discipline Specific): 0

Total Adjective-Noun Combination: 14

Adjective-Noun Combination (Discipline Specific): 4

Adjective-Noun Combination (Non-Discipline Specific): 10

Combination of Both: 1 (DS) 0 (NDS)

12. Sharif, M. I., Mehmood, M., Uddin, M. P., Siddique, K., Akhtar, Z., & Waheed, S. (2024). Federated learning for analysis of medical images: A survey. Journal of Computer Science, 20(12), 1610–1621. https://doi.org/10.3844/jcssp.2024.1610.1621

Machine learning models trained in **medical imaging** can help in the early detection, diagnosis, and prognosis of the disease. However, it confronts two major obstacles: **deep learning models** require access to a substantial amount of imaging data**,** which is a hard constraint, and the **patient data** is private and sensitive, so it cannot be shared like 1 other imaging data in

computer vision. Federated Learning (FL) offers an alternative by deploying many training models in a decentralized way. In recent years, various techniques that leverage FL for **disease diagnosis** have been introduced. Existing **survey articles** have analyzed and collated research about the use of FL in general. However, the particular component of medical imaging is ignored. The motivation behind this **survey paper** is to fill up the **research gap** by providing a comprehensive survey of **FL techniques** for medical imaging and various ways in which FL is employed to provide secure, accessible, and collaborative deep **learning models** for the medical imaging **research community**.

Word Count: 163

Total Combinations: 25

Total Noun-Noun Combination: 10

Noun-Noun Combination (Discipline Specific): 5

Noun-Noun Combination (Non-Discipline Specific): 5

Total Adjective-Noun Combination: 12

Adjective-Noun Combination (Discipline Specific): 1

Adjective-Noun Combination (Non-Discipline Specific): 11

Combination of Both: 1 (DS) 2 (NDS)

13. Shah, S., & Patel, S. (2025). A comprehensive survey on fake news detection using machine learning. Journal of Computer Science, 21(4), 982–990. https://doi.org/10.3844/jcssp.2025.982.990

In the age of **information data surplus** and **social media impact**, the propagation of fake news has developed a significant societal concern. This challenge requires robust tools and methodologies for addressing, with **machine learning** emerging as a promising approach. The paper reviews different **machine learning techniques** in fake news detection, including **supervised, unsupervised and semi-supervised methods**. **Supervised methods** utilize labelled datasets to train models to discriminate between fake and legitimate **news articles**. **Unsupervised Learning methods**, on the other hand, rely on clustering as well as **anomaly detection** to identify suspicious patterns in **information data**. **Semi-supervised techniques** associate elements of both **supervised and unsupervised learning** techniques for leveraging limited labelled data successfully. Moreover, the manuscript examines **feature extraction techniques**, including **Natural Language Processing** (NLP) techniques like bag-of-words, **word embedding** and syntactic parsing. It also discusses the importance of incorporating contextual information, such as **source credibility** and **social network dynamics**, into the detection process. The paper addresses the **evaluation metrics** normally used to evaluate the performance of fake news detection models, such as accuracy percentage, precision, recall and F1-score. It highlights the need for robust evaluation frameworks to ensure the reliability and generalizability of the fake news detection system.

Word Count: 197

Total Combinations: 33

Total Noun-Noun Combination: 18

Noun-Noun Combination (Discipline Specific): 11

Noun-Noun Combination (Non-Discipline Specific): 7

Total Adjective-Noun Combination: 15

Adjective-Noun Combination (Discipline Specific): 5

Adjective-Noun Combination (Non-Discipline Specific): 10

Combination of Both: 0 (DS) 0 (NDS)

14. Velmurugan, T., & Santhanam, T. (2010). Computational complexity between k-means and k-medoids clustering algorithms for normal and uniform distributions of data points. Journal of Computer Science, 6(3), 363–368. https://doi.org/10.3844/jcssp.2010.363.368

Problem statement: Clustering is one of the most important research areas in the field of **data mining**. Clustering means creating groups of objects based on their features in such a way that the objects belonging to the same groups are similar and those belonging to different groups are dissimilar. Clustering is an unsupervised learning technique. The main advantage of clustering is that interesting patterns and structures can be found directly from very large **data sets** with little or none of the background knowledge. **Clustering algorithms** can be applied in many domains. Approach: In this research, the most representative algorithms K-Means and K-Medoids were examined and analyzed based on their basic approach. The best algorithm in each category was found out based on their performance. The **input data** points are generated by two ways, one by using normal distribution and another by applying uniform distribution. Results: The randomly distributed **data points** were taken as input to these algorithms and clusters are found out for each algorithm. The algorithms were implemented using **JAVA language** and the performance was analyzed based on their **clustering quality**. The **execution time** for the algorithms in each category was compared for different runs. The accuracy of the algorithm was investigated during different execution of the program on the **input data points**. Conclusion: The average time taken by **K-Means algorithm** is greater than the time taken by **K-Medoids algorithm** for both the case of normal and uniform distributions. The results proved to be satisfactory.

Word Count: 246

Total Combinations: 28

Total Noun-Noun Combination: 10

Noun-Noun Combination (Discipline Specific): 8

Noun-Noun Combination (Non-Discipline Specific): 2

Total Adjective-Noun Combination: 16

Adjective-Noun Combination (Discipline Specific): 2

Adjective-Noun Combination (Non-Discipline Specific): 14

Combination of Both: 2 (DS) 0 (NDS)

15. Goon, S., Pal, D., & Dihidar, S. (2022). Primary color based numerous image creation in visual cryptography with the help of grasshopper algorithm, artificial neural network and elliptic curve cryptography. *Journal of Computer Science, 18*(5), 322–338. https://doi.org/10.3844/jcssp.2022.322.338

The two primary restrictions of **visual cryptography methods**, contract, and security are well-known. **Visual cryptography** is considered secure if no share reveals information about the secure picture and this image is unreachable when shares are combined. Recursion can be used in a **visual cryptography method** to ensure the integrity of both arguments. Sometimes, the many shares are generated illogically in such a manner that they sometimes reveal the actual image. The model proposed in the paper has been proved beneficial in reducing the computation, noise, or distortion in the recreated image and helps in providing amended contracts. The suggested scheme uses unique **pixel patterns** to create a contract for the decrypted image. The contract for the **recursion visual cryptography scheme** is alterable from the white **pixel pattern** to new **pixel patterns**. The **elliptic curve cryptography approach** is used to ensure the privacy and security of the image. This study has taken a **four-share mechanism** in which each component is divided into four shares for each **color segment.** The proposed algorithm uses the **Grasshopper algorithm** to select the bit's positions where the encryption must be done. **Neural Networks** have been used as a cross validator in the proposed **work model**. A new fitness function is designed for the **Grasshopper algorithm**. The proposed algorithm is compared based on **Mean Square Error** and **Peak Signal** to **Noise Ratio.**

Word Count: 225

Total Combinations: 27

Total Noun-Noun Combination: 13

Noun-Noun Combination (Discipline Specific): 10

Noun-Noun Combination (Non-Discipline Specific): 3

Total Adjective-Noun Combination: 10

Adjective-Noun Combination (Discipline Specific): 3

Adjective-Noun Combination (Non-Discipline Specific): 7

Combination of Both: 3 (DS) 1 (NDS)

Abstract Data Combined Table

Abstract 1	**Abstract 2**	**Abstract 3**	**Abstract 4**
Word Count: 208	**Word Count: 233**	**Word Count: 256**	**Word Count:253**
Total Combinations: 29	**Total Combinations: 25**	**Total Combinations: 39**	**Total Combinations: 43**
Total Noun-Noun Combination: 18	**Total Noun-Noun Combination: 5**	**Total Noun-Noun Combination: 10**	**Total Noun-Noun Combination: 11**
Noun-Noun Combination (Discipline Specific): 14	Noun-Noun Combination (Discipline Specific): 5	Noun-Noun Combination (Discipline Specific): 3	Noun-Noun Combination (Discipline Specific): 9
Noun-Noun Combination (Non-Discipline Specific): 4	Noun-Noun Combination (Non-Discipline Specific): 0	Noun-Noun Combination (Non-Discipline Specific): 7	Noun-Noun Combination (Non-Discipline Specific): 2
Total Adjective-Noun Combination: 10	**Total Adjective-Noun Combination: 16**	**Total Adjective-Noun Combination: 24**	**Total Adjective-Noun Combination: 16**
Adjective-Noun Combination (Discipline Specific): 0	Adjective-Noun Combination (Discipline Specific): 1	Adjective-Noun Combination (Discipline Specific): 4	Adjective-Noun Combination (Discipline Specific): 2
Adjective-Noun Combination (Non-Discipline Specific): 10	Adjective-Noun Combination (Non-Discipline Specific): 15	Adjective-Noun Combination (Non-Discipline Specific): 20	Adjective-Noun Combination (Non-Discipline Specific): 14
Combination of Both: 1 (DS) 0 (NDS)	**Combination of Both: 3 (DS) 1(NDS)**	**Combination of Both: 2 (DS) 3 (NDS)**	**Combination of Both: 16 (DS) 0 (NDS)**
Abstract 5	**Abstract 6**	**Abstract 7**	**Abstract 8**
Word Count: 138	**Word Count: 201**	**Word Count: 150**	**Word Count: 234**
Total Combinations: 18	**Total Combinations: 14**	**Total Combination: 23**	**Total Combination: 32**
Total Noun-Noun Combination: 6	**Total Noun-Noun Combination: 3**	**Total Noun-Noun Combination: 14**	**Total Noun-Noun Combination: 12**

Noun-Noun Combination (Discipline Specific): 4	Noun-Noun Combination (Discipline Specific): 2	Noun-Noun Combination (Discipline Specific): 3	Noun-Noun Combination (Discipline Specific): 6
Noun-Noun Combination (Non-Discipline Specific): 2	Noun-Noun Combination (Non-Discipline Specific): 1	Noun-Noun Combination (Non-Discipline Specific): 11	Noun-Noun Combination (Non-Discipline Specific): 6
Total Adjective-Noun Combination: 7	**Total Adjective-Noun Combination: 8**	**Total Adjective-Noun Combination: 7**	**Total Adjective-Noun Combination: 14**
Adjective-Noun Combination (Discipline Specific): 0	Adjective-Noun Combination (Discipline Specific): 8	Adjective-Noun Combination (Discipline Specific): 1	Adjective-Noun Combination (Discipline Specific): 5
Adjective-Noun Combination (Non-Discipline Specific): 7	Adjective-Noun Combination (Non-Discipline Specific): 0	Adjective-Noun Combination (Non-Discipline Specific): 6	Adjective-Noun Combination (Non-Discipline Specific): 9
Combination of Both: 0 (DS) 5 (NDS)	**Combination of Both: 3 (DS) 0 (NDS)**	**Combination of Both: 1 (DS) 1 (NDS)**	**Combination of Both: 3 (DS) 3 (NDS)**
Abstract 9	**Abstract 10**	**Abstract 11**	**Abstract 12**
Word Count: 163	**Word Count: 228**	**Word Count: 209**	**Word Count: 163**
Total Combinations: 15	**Total Combinations: 45**	**Total Combinations: 27**	**Total Combinations: 25**
Total Noun-Noun Combination: 6	**Total Noun-Noun Combination: 24**	**Total Noun-Noun Combination: 12**	**Total Noun-Noun Combination: 10**
Noun-Noun Combination (Discipline Specific): 3	Noun-Noun Combination (Discipline Specific): 9	Noun-Noun Combination (Discipline Specific): 12	Noun-Noun Combination (Discipline Specific): 5
Noun-Noun Combination (Non-Discipline Specific): 3	Noun-Noun Combination (Non-Discipline Specific): 15	Noun-Noun Combination (Non-Discipline Specific): 0	Noun-Noun Combination (Non-Discipline Specific): 5
Total Adjective-Noun Combination: 9	**Total Adjective-Noun Combination: 20**	**Total Adjective-Noun Combination: 14**	**Total Adjective-Noun Combination: 12**
Adjective-Noun Combination	Adjective-Noun Combination	Adjective-Noun Combination	Adjective-Noun Combination

(Discipline Specific): 0	(Discipline Specific): 3	(Discipline Specific): 4	(Discipline Specific): 1
Adjective-Noun Combination (Non-Discipline Specific): 9	Adjective-Noun Combination (Non-Discipline Specific): 17	Adjective-Noun Combination (Non-Discipline Specific): 10	Adjective-Noun Combination (Non-Discipline Specific): 11
Combination of Both: 0 (DS) 0 (NDS)	**Combination of Both: 1 (DS) 0 (NDS)**	**Combination of Both: 1 (DS) 0 (NDS)**	**Combination of Both: 1 (DS) 2 (NDS)**
Abstract 13	**Abstract 14**	**Abstract 15**	
Word Count: 197	**Word Count: 246**	**Word Count: 225**	
Total Combinations: 33	**Total Combinations: 28**	**Total Combinations: 27**	
Total Noun-Noun Combination: 18	**Total Noun-Noun Combination: 10**	**Total Noun-Noun Combination: 13**	
Noun-Noun Combination (Discipline Specific): 11	Noun-Noun Combination (Discipline Specific): 8	Noun-Noun Combination (Discipline Specific): 10	
Noun-Noun Combination (Non-Discipline Specific): 7	Noun-Noun Combination (Non-Discipline Specific): 2	Noun-Noun Combination (Non-Discipline Specific): 3	
Total Adjective-Noun Combination: 15	**Total Adjective-Noun Combination: 16**	**Total Adjective-Noun Combination: 10**	
Adjective-Noun Combination (Discipline Specific): 5	Adjective-Noun Combination (Discipline Specific): 2	Adjective-Noun Combination (Discipline Specific): 3	
Adjective-Noun Combination (Non-Discipline Specific): 10	Adjective-Noun Combination (Non-Discipline Specific): 14	Adjective-Noun Combination (Non-Discipline Specific): 7	
Combination of Both: 0 (DS) 0 (NDS)	**Combination of Both: 2 (DS) 0 (NDS)**	**Combination of Both: 3 (DS) 1 (NDS)**	